WASP-10b

R-0659

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_251213` · created 2026-07-27T17:30:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461022.687 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1749 +/- 0.0099
Transit depth (Rp/Rs)^2	3.06 +/- 0.35 [%]
Orbital Inclination (inc)	87.71 +/- 0.66
Airmass coefficient 1 (a1)	1.006 +/- 0.013
Airmass coefficient 2 (a2)	-0.0038 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#2 - [241, 298]
Optimal Aperture	3.68
Optimal Annulus	8.11
Transit Duration (day)	0.088 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1749 ± 0.0099 vs 0.1592 (deviation 1.59σ)
Duration fit vs published	0.088 d vs 0.0946 d (deviation 1.32σ)
Mid-time O–C	0.4 min
Depth SNR	17.7
Residual scatter	1.25 %

Light curve

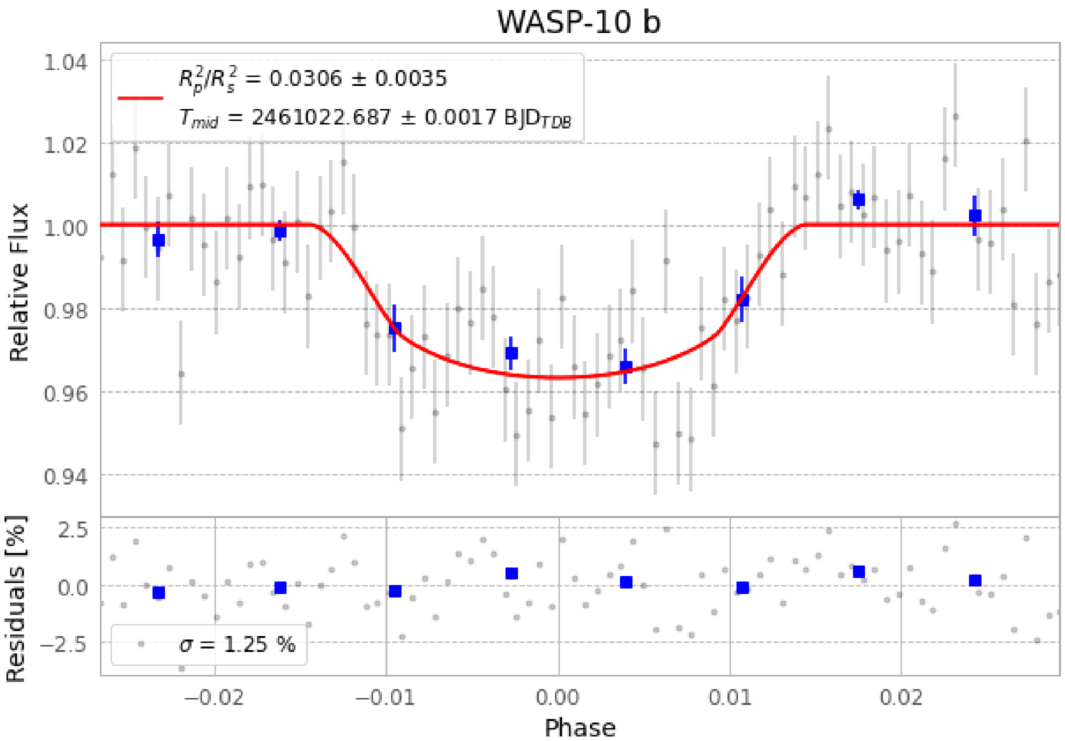

Figure 1 — FinalLightCurve_WASP-10 b_2025-12-12.png (SHA-256 3937a3c5a276e21a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 343], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c28bdd58124fae44…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e`

Data provenance

84 FITS frames (55 MB), `WASP-10251213022716.FITS` → `WASP-10251213063616.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-251213.log (ec4b97bb07e9…), FinalLightCurve_WASP-10 b_2025-12-12.png (3937a3c5a276…), FinalLightCurve_WASP-10 b_2025-12-12.csv (3ffadde8c436…)

Compute resources

Wall time 395.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-27 17:30Z.